Shazli Mekrani

github.com/ShazliMek | linkedin.com/in/shazlim | mekranishazli@gmail.com | +1 (519)7018419 | Portfolio

Education

Western University — London, ON
Candidate for Honors Specialization in Computer Science, Minor in Software Engineering — *Expected Graduation 2027*

- Dean's Honour List 2024/25 **- 3.9 GPA**

Experience

Data Engineer & Researcher — September 2026 – Present
Ivey Buissness School & Interpid Data Solutions — *London, ON, CA*

- Building a data platform to uncover hidden delays in Canadian medical device access, benchmarking 3 regulators (Health Canada, FDA, EU) across Class II-IV devices and 13 provinces and territories

Founding Technical Engineer — July 2026 – Present
Homni | Office of the CTO — *Toronto, ON, CA*

- Built Homni, a **full-stack CRM using React, TypeScript, Express, and PostgreSQL** to centralize client records, sales pipelines, rentals, and renovation workflows for mobility equipment dealers.
- Developed an AI intake pipeline using **Whisper and Claude to transcribe calls**, extract structured client information, and flag ambiguous updates for human review.
- Integrated **Twilio voice** and SMS alongside **Google and Microsoft email services**, linking customer communications to shared conversation histories.
- Implemented **authenticated API** access with organization-scoped sessions and role context to protect customer data across dealer accounts.

Software Engineer Intern — January 2026 – August 2026
Lumentum | Software Engineering Tools and Infrastructure Team — *Ottawa, ON, CA*

- Developed production automation software supporting **OCS manufacturing workflows**, improving station reliability, test repeatability, and operator efficiency.
- Created hardware drivers for **pressure monitoring systems** and **stepper motor controllers**, automating production test workflows and reducing cycle time by **70%**.
- Supported migration from **SVN to GitHub**, improving source control practices, collaboration, maintainability, and CI/CD readiness for internal software projects.
- Authored technical documentation and incorporated developer feedback to improve maintainability, onboarding, and feature adoption across internal engineering tools.

Software Engineering Intern — September 2025 – December 2025
Lumentum | Process Automation Software Team — *Ottawa, ON, CA*

- Developed and maintained internal **.NET libraries, databases, and automation frameworks** supporting production engineering workflows and **OCS process automation**.
- Automated **ETL pipelines** for production test data used by **10+ cross-functional teams**, improving data availability and operational efficiency by **20%**.
- Built internal tooling, automation scripts, and release pipelines to improve software infrastructure reliability, streamline manufacturing workflows, and increase developer productivity.

AI/ML Intern — May 2025 – August 2025
DIWO — *Northville, MI, USA*

- Implemented a **Python MCP** server for **async message routing**, cutting latency by **15%**.
- **Containerized server and ML modules** with **Docker** and **AWS ECS** for consistent deployments.
- **Authored API documentation** and coordinated inference endpoint integration across teams.

Projects

AI Mock Interview Platform | *Next.js, Gemini API, Firebase, Typescript, VAPI, MongoDB* | Code — Jun 2025

- Developed an **AI-powered** mock interview platform using **Firebase Auth, Firestore, and Google Generative AI API** and **Vapi** for voice interactions and dynamic question generation.
- End-to-end encryption with **UUID-based session tracking**, secure **API** endpoints, and **GDPR**-compliant data handling protocols.

StockOverflow | *Python, JS, HTML/CSS, Jupyter* | Code — May 2024

- Built a fraud detection and stock prediction model using **XGBoost** and **Random Forest**, elevating accuracy from **93% to 98%** via K-fold cross-validation.
- Implemented **NEAR blockchain** integration with **Rust smart contracts**, increasing data integrity by **65%** and cutting transaction times by **300 ms/request**.

Technical Skills

Languages: Python, HTML, CSS, Java, Perl, JavaScript, Tailwind CSS, TypeScript, C, C++, C#, SQL/SQLite, R
Frameworks: .NET Core, WPF, ReactJS, Vue.js, Node.js, Express.js, Django, REST APIs, FastAPI, Pandas, OpenCV, sklearn, TensorFlow, matplotlib, seaborn, Java Swing, JavaFX, Qt, CMake, QMake, Next.js
Developer Tools: Azure, Git, GitHub, Docker, GCP, Visual Studio, Jira, Linux, DataBricks, VAPI, Vercel, Firebase